

Nowell Hollier Phelps

nowellphelps.github.io | nphelps2@jh.edu | Last updated: September 1st, 2026

Education

Johns Hopkins University *Doctor of Philosophy (Ph.D.), Biostatistics* 2024 – 2029 (expected)

- Coursework in statistical theory, measure-theoretic probability, biostatistical methods and statistical computing
- Advised by Dr Abhi Datta

University of Oxford *Master of Mathematics (MMath), Mathematics* 2017 - 2021

- Part C (Master's component): *Distinction*; Parts A&B (Undergraduate component): *Second Class, Division One*
- Dissertation "A coevolutionary voter model for opinion dynamics on hypergraphs", advised by Dr Renaud Lambiotte

Research Experience

African Institute for Mathematical Sciences Research and Innovation Centre *Visiting Researcher* March - July 2024

Research visit at AIMS RIC in Kigali, Rwanda, collaborating on developing a database for climate and health research.

Imperial College London *Research Assistant in Epidemiology and Population Health* March 2022 - July 2024

RA in the NCD Risk Factor Collaboration, WHO Collaborating Centre on NCD Surveillance, Epidemiology and Modelling, supervised by Dr Majid Ezzati. Led analysis updating the WHO's national underweight and obesity prevalence estimates.

Big Data Institute, University of Oxford *Summer Studentship* July - September 2021

Eight-week research internship investigating spatial risk of schistosomiasis, supervised by Dr Goylette Chami.

Mathematical Institute, University of Oxford *Undergraduate Research Internship* August - September 2020

Four-week research internship on social network effects on microfinance adoption, supervised by Dr Neave O'Clery.

Publications

Statistical methodology

1. Victor PF Lhoste, Yefeng Fan, James E Bennett, Sarah Filippi, Christopher J Paciorek, Junyang Wang, Bin Zhou, Theo Rashid, **Nowell H Phelps**, Majid Ezzati. "Hierarchical Bayesian mixture model for subgroup-specific cardiometabolic phenotypes", *in press at Journal of the Royal Statistical Society Series A* (2026)

Scientific application – global health

2. **[Co-first author]** NCD Risk Factor Collaboration (NCD-RisC). "Obesity rise plateaus in developed nations and accelerates in developing nations", *Nature* (2026)
 - Press coverage in the [Independent](#), [Guardian](#), [Le Monde](#), and [more](#)
3. NCD Risk Factor Collaboration (NCD-RisC). "Worldwide trends in diabetes prevalence and treatment from 1990 to 2022: a pooled analysis of 1108 population-representative studies with 141 million participants", *The Lancet* (2024)
4. NCD Risk Factor Collaboration (NCD-RisC). "General and abdominal adiposity and hypertension in eight world regions: a pooled analysis of 837 population-based studies with 7.5 million participants", *The Lancet* (2024)
5. **[Co-first author]** NCD Risk Factor Collaboration (NCD-RisC) "Worldwide trends in the burden of underweight and obesity from 1990 to 2022: a pooled analysis of 3663 population-representative studies with 222 million school-aged children, adolescents and adults", *The Lancet* (2024)
 - >3,000 citations (as of August 2026), with coverage in the [BBC](#), [Economist](#), [Guardian](#), [CNN](#), and [more](#)
6. NCD Risk Factor Collaboration (NCD-RisC). "Global variation in diabetes diagnosis and prevalence based on fasting glucose and hemoglobin A1c", *Nature Medicine* (2023)
7. NCD Risk Factor Collaboration (NCD-RisC). "Diminishing benefits of urban living for children and adolescents' growth and development", *Nature* (2023)

* denotes equal contribution (co-lead or co-senior authorship)

Teaching

Department of Biostatistics, Johns Hopkins University Teaching Assistant

Statistical Methods in Public Health I-IV

September 2026 – May 2027

- Master's level introductory biostatistics course

Advanced Methods in Biostatistics III-IV

January - May 2026

- PhD-level classes on biostatistical methods. Weekly office hours; bi-weekly lab session; grading.

Advanced Probability Theory I-II

September - December 2025

- PhD-level classes on measure-theoretic probability. Weekly office hours; grading; class on sequences taught 4 September 2025; review class taught 15 October 2025. Provided extra support in linear algebra foundations.

Department of Biostatistics, Johns Hopkins University Tutoring

May - June 2026

- Tutoring for first-year PhD student on *Advanced Methods in Biostatistics I-IV* PhD qualifying exam

Presentations, posters and workshops

Invited seminars and panels

1. "The obesity pandemic: trends and perspectives" *Webinar, Zurich Heart House* (April 30th, 2026)
2. "Understanding the global burden of obesity and the implications for cardiovascular health" *Lancet panel on obesity and cardiovascular health outcomes, American Heart Association Scientific Sessions* (November 26th, 2024)
3. "MCMC for a large Bayesian hierarchical model" *Statistics Seminar Group, University of Kent* (October 12th, 2023)

Conference posters

1. **Nowell H Phelps**, ..., Abhirup Datta, Sarath Raju "Outdoor air pollutant exposure is associated with accelerated lung function decline in a high-risk HIV cohort" *American Thoracic Society International Conference* (May 20th, 2026)

Departmental presentations

1. "Modelling trends in population-level obesity" *BEST seminar* (April 22nd, 2026)
2. "Random walk models for population genetics" *Advanced Probability Theory IV class* (April 15th, 2026)
3. "Bayesian hierarchical modelling of population-level obesity" *WIT working group* (January 24th, 2025)

Service and outreach

Organizer, Bayesian Learning & Spatio-Temporal Modelling (BLAST) Working Group, JHU Biostatistics

August 2026 – Present

Co-lead, Tea-time JHU Biostatistics

August 2025 – May 2026

PhD Student Mentor JHU Biostatistics

May 2025 – Present

Co-lead, Curriculum Committee JHU Biostatistics Student Organization

May 2025 – Present

- Designed visualization of the PhD program core curriculum, integrated into official department materials; lead information sessions on PhD qualifying and oral exams; disseminate student feedback to faculty

Site Manager Opportunity Oxford Residential Program, University of Oxford

September 2020 & September 2021

Ambassador UNIQ Summer Program, University of Oxford

July – August 2020

Team Guide International Mathematics Olympiad, Bath

July 2019

Mathematics Subject Representative Jesus College JCR, University of Oxford

October 2018 – July 2019

Junior Members' Scholarship Representative Jesus College JCR, University of Oxford

January – December 2018

Awards

Gordis Teaching Fellowship Johns Hopkins University (\$8000)

September 2026

- Awarded to develop and teach course titled "Creating and critiquing data visualizations for public health" to upper-level undergraduates in JHU's Public Health Studies Program. Course to be scheduled in fall 2027.

College Prize Jesus College, University of Oxford. For distinction in Part C of MMath

August 2021

Summer Studentship Big Data Institute, University of Oxford (£3000)

July 2021

Open Exhibition Jesus College, University of Oxford For first class performance in Part B of MMath

August 2020

College Prize Jesus College, University of Oxford For excellent performance in internal college examinations

January 2018

References available upon request